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The equity-environmental sustainability nexus in autonomous vehicles: An integrated framework for policy and planning

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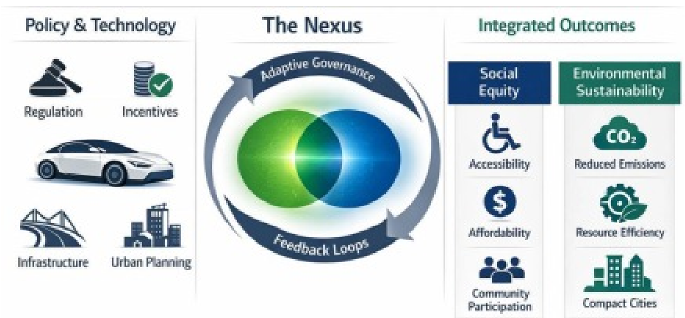
Highlights

- Develops an integrated equity–environmental sustainability framework for autonomous vehicle deployment.
- Identifies critical policy, planning, and assessment gaps arising from siloed AV research.
- Synthesizes equity dimensions, sustainability objectives, and policy levers within a unified conceptual model.
- Emphasizes proactive planning, community engagement, and adaptive governance for AV systems.
- Proposes an application pathway to support iterative, data-driven, and context-sensitive policymaking.

Abstract

Autonomous Vehicles (AVs) present a dual challenge: their promise of advancing environmental sustainability through efficiency gains is counterbalanced by significant social equity concerns, such as job displacement and unequal access for vulnerable populations. Despite these opportunities and risks, there remains a critical gap in integrated frameworks that optimize both social equity and environmental sustainability in AV deployment. To address this, we conducted a comprehensive literature review, identifying key voids in AV policy, community impacts, behavioral acceptance, and integrated assessment methodologies. Building on these findings, we developed a novel conceptual framework for equitable and sustainable AV deployment. This framework emphasizes the dynamic interplay between AV technology, policy interventions, and their social and environmental outcomes. It is grounded in proactive planning, meaningful community engagement, adaptive governance, and interdisciplinary collaboration, and is designed to be adaptable across diverse deployment contexts. Results provide both theoretical insights and practical recommendations, including an application pathway for iterative policy adaptation. These contributions offer a roadmap for policymakers and planners to guide AV deployment towards mutually reinforcing equitable and sustainable outcomes, ensuring AVs contribute positively to a future that is both green and fair.

Graphical abstract



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Keywords

Autonomous vehicles; Environmental sustainability; Social equity; Transportation policy; Urban planning; Integrated framework; Research gaps; Mobility justice

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Introduction

Autonomous Vehicles (AVs) represent a transformative frontier in transportation, promising to revolutionize urban mobility, logistics, and personal travel. Proponents often highlight their potential to enhance environmental sustainability through optimized traffic flow, reduced congestion, increased vehicle utilization, and the accelerated adoption of electric powertrains (Emory, Douma, & Cao, 2022). These advancements are envisioned to lead to significant reductions in greenhouse gas emissions; energy consumption; and urban land dedicated to parking (Kopelias, Demiridi, Vogiatzis, Skabardonis, & Zafiropoulou, 2020). However; alongside these promising environmental benefits; the widespread deployment of AVs introduces complex challenges; particularly concerning social equity. Concerns have emerged regarding potential job displacement for professional drivers; unequal access to AV services; and the exacerbation of existing transportation disparities for vulnerable populations (Fatima, Hsiu Lee, & Dannenberg, 2024). This article posits that the relationship between the environmental sustainability and social equity impacts of AVs is not merely coincidental but deeply intertwined; forming a critical equity-environmental sustainability nexus. A failure to address this nexus comprehensively risks undermining the very goals of sustainable development. For instance; while AVs could theoretically reduce emissions; if their deployment leads to increased Vehicle Miles Traveled (VMT) due to cheaper; more convenient personal travel; or if they disproportionately benefit affluent communities while burdening marginalized ones with increased traffic or reduced public transit options; the net environmental and social outcomes could be negative (Freemark, Stacy, Fiol, Morales-Burnett, & Weng, 2022).

The overarching research gap addressed in this article is the pervasive lack of integrated policy, planning, and assessment frameworks that simultaneously consider and optimize for both social equity and environmental sustainability in the context of AV deployment. Existing research and policy often treat these two critical dimensions in isolation, leading to fragmented understanding and potentially conflicting outcomes. This siloed approach prevents the development of holistic strategies necessary for a truly sustainable and just autonomous mobility future (Bahman, Abahussain, Khan, Mahmood, & Shaker, 2025).

While AVs represent a transformative frontier in transportation, their deployment is shaped by a complex ecosystem that extends beyond equity and environmental sustainability. Key factors including governance arrangements, safety assurance, technology readiness, market structures, and public acceptance, play critical roles in determining AV outcomes across diverse operational and governance dimensions. This article intentionally narrows its focus to the equity–environmental sustainability nexus, acknowledging that other dimensions are essential but outside the scope of the present analysis. The rationale for this boundary is to provide theoretical clarity and depth, while recognizing the broader tensions and uncertainties that characterize AV development and deployment. A balanced overview of these ecosystem challenges is provided below, before the discussion narrows to the integrated equity-environment sustainability lens.

This paper focuses specifically on autonomous passenger transport in urban contexts. This focus reflects both the dominant emphasis of the existing literature and the concentration of current policy experimentation and pilot deployments in cities. While autonomous systems are also emerging in freight and logistics, these domains raise distinct equity and environmental sustainability challenges that fall outside the scope of the present analysis. Narrowing the scope in this way allows for greater conceptual clarity and analytical depth in examining the equity–environmental sustainability nexus.

This article aims to address this gap by:

1. Critically reviewing the existing literature on the environmental and social equity impacts of AVs, highlighting the disconnects and specific research voids.

2. Proposing a novel conceptual framework that integrates equity and sustainability considerations into AV policy and planning.

3. Providing concrete policy and planning implications for stakeholders to guide AV deployment towards mutually reinforcing equitable and sustainable outcomes.

By exploring this nexus, this article seeks to make a significant contribution to literature, offering both theoretical insights and practical recommendations for navigating the complex interplay between technological advancement, environmental stewardship, and social justice in the future of transportation.

For clarity and consistency, the following definitions are adopted: *Social equity* refers to the fair distribution of benefits and burdens across all societal groups, ensuring equal access to opportunities and resources regardless of socio-economic status. *Autonomous Vehicles (AVs)* refers to all forms of self-driving vehicles and associated services, including shared and on-demand AVs operating within urban and regional contexts. Throughout the paper, *sustainability* is understood as the simultaneous pursuit of environmental, social, and economic objectives, aligning with the widely recognized three-pillar framework.

The remainder of this paper is organized as follows: Section 2 reviews the relevant literature and identifies key research gaps. Section 3 introduces the proposed integrated framework for equitable and sustainable AV deployment, detailing its core components and policy levers. Section 4 discusses the implications of the framework, including practical pathways for implementation and mechanisms for addressing trade-offs. Finally, Section 5 presents the conclusions, highlighting contributions, limitations, and directions for future research.

Section snippets

Literature review

Sustainability is commonly conceptualized as encompassing three interrelated pillars: environmental, social, and economic (Elkington, 1997). These pillars collectively define the scope of sustainable development, with each dimension contributing distinct yet interconnected objectives: environmental protection, social equity, and economic viability. In the context of AV deployment, these elements often interact in complex ways, shaping both policy priorities and practical outcomes. While this …

Towards an integrated framework for equitable and sustainable AV deployment

This paper adopts a conceptual and interpretive approach rather than a systematic or integrative review. Articles were selected to reflect influential policy debates, empirical insights, and conceptual contributions relating to autonomous passenger transport, equity, and environmental sustainability. The analysis involved thematic synthesis across these strands to identify recurring gaps, tensions, and integration challenges that informed the development of the proposed framework. …

Discussion and implications

The proposed integrated framework for equitable and sustainable AV deployment directly addresses the research gaps identified in Section 2.4 by providing a structured approach to bridge the current disconnects between equity and sustainability considerations. By emphasizing the interdependencies between AV technology, policy interventions, and their social and environmental outcomes, the framework encourages a holistic perspective that moves beyond siloed analyses. Specifically, the framework …

Conclusion

This article has explored the critical equity-environmental sustainability nexus in urban autonomous passenger transport, arguing that an integrated policy and planning approach is essential for realizing the potential benefits of AVs while safeguarding environmental outcomes and social equity. We have demonstrated that current research and policy often treat equity and sustainability as separate concerns, leading to fragmented understanding and the risk of unintended negative consequences. The …

Use of Artificial Intelligence (AI)

This manuscript was prepared by the authors with the assistance of AI-based writing support tools for language and editorial clarity. All analysis, interpretation, and conclusions are original. …

CRediT authorship contribution statement

Nahed Bahman: Visualization, Project administration, Investigation, Formal analysis, Data curation, Writing – review & editing, Writing – original draft.

Naser Naser: Visualization, Conceptualization, Writing – original draft. **Tariq Mahmood:** Supervision, Writing – review & editing, Writing – original draft. …

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Declaration of competing interest

The authors declare that there is no conflict of interest relevant to the content of this manuscript. …

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